Chapter 4

Congruent Triangles

Unit 2 — Triangles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Congruent polygons	All corresponding sides and angles are congruent
CPCTC	Corresponding Parts of Congruent Triangles are Congruent
Included angle	The angle between two given sides
Included side	The side between two given angles
Isosceles triangle	At least two congruent sides
Equilateral triangle	All three sides congruent
Corollary	A theorem that follows directly from another

Theorem 4-1: Triangle Angle Sum

> The sum of the measures of the angles of a triangle is **180°**.

$$m\angle A + m\angle B + m\angle C = 180^\circ$$

Example: If $m\angle A = 50^\circ$ and $m\angle B = 70^\circ$, then $m\angle C = 60^\circ$.

Theorem 4-2: Exterior Angle Theorem

An **exterior angle** of a triangle equals the sum of the two **non-adjacent interior angles**.

$$m\angle 4 = m\angle 1 + m\angle 2$$

```
        A
       /|
      / |
     /  |
    B───C──→ (exterior angle at C)
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Corollary 4-1: Third Angle Corollary

> If two angles of one triangle are congruent to two angles of another triangle, then the **third angles** are also congruent.

$$\angle A \cong \angle D \text{ and } \angle B \cong \angle E \Rightarrow \angle C \cong \angle F$$

Postulate 4-1: SSS (Side-Side-Side) Congruence

> If **three sides** of one triangle are congruent to three sides of another, the triangles are **congruent**.

$$AB = DE,\ BC = EF,\ AC = DF \Rightarrow \triangle ABC \cong \triangle DEF$$

No angle information needed.

Postulate 4-2: SAS (Side-Angle-Side) Congruence

> If **two sides** and the **included angle** of one triangle are congruent to those of another, the triangles are **congruent**.

$$AB = DE,\ \angle B \cong \angle E,\ BC = EF \Rightarrow \triangle ABC \cong \triangle DEF$$

The angle must be between the two sides.

Postulate 4-3: ASA (Angle-Side-Angle) Congruence

> If **two angles** and the **included side** of one triangle are congruent to those of another, the triangles are **congruent**.

$$\angle A \cong \angle D,\ AB = DE,\ \angle B \cong \angle E \Rightarrow \triangle ABC \cong \triangle DEF$$

The side must be between the two angles.

Theorem 4-3: AAS (Angle-Angle-Side) Congruence

> If **two angles** and a **non-included side** of one triangle are congruent to those of another, the triangles are **congruent**.

$$\angle A \cong \angle D,\ \angle B \cong \angle E,\ BC = EF \Rightarrow \triangle ABC \cong \triangle DEF$$

AAS is a theorem (proved), not a postulate.

Theorem 4-4: HL (Hypotenuse-Leg) Congruence

> For **right triangles**: if the **hypotenuse** and one **leg** are congruent, the triangles are **congruent**.

$$\text{Right } \triangle ABC \cong \text{Right } \triangle DEF \text{ if hyp and one leg are } \cong$$

Only applies to right triangles.

Congruence Shortcut Summary

Shortcut	Type	Notes
SSS	Postulate	3 sides
SAS	Postulate	2 sides + included ∠
ASA	Postulate	2 ∠s + included side
AAS	Theorem	2 ∠s + non-included side
HL	Theorem	Right triangles only
~~AAA~~	✗ NOT valid	Proves similarity, not congruence
~~SSA~~	✗ NOT valid	"Ambiguous case"

Theorem 4-5: Isosceles Triangle Theorem

> If two sides of a triangle are congruent, then the **angles opposite** those sides are congruent.

$$\overline{AB} \cong \overline{AC} \Rightarrow \angle B \cong \angle C$$

(Base angles of an isosceles triangle are congruent.)

Theorem 4-6: Converse of Isosceles Triangle Theorem

> If two angles of a triangle are congruent, then the **sides opposite** those angles are congruent.

$$\angle B \cong \angle C \Rightarrow \overline{AB} \cong \overline{AC}$$

Corollary 4-2: Equilateral Triangle

A triangle is **equilateral** if and only if it is **equiangular**.

$$\overline{AB} \cong \overline{BC} \cong \overline{AC} \iff \angle A = \angle B = \angle C = 60^\circ$$

Example — Proving Congruence

Given: M is the midpoint of $\overline{AB}$; $\angle ACM \cong \angle BCM$

Prove: $\triangle ACM \cong \triangle BCM$

Statement	Reason
$AM = BM$	Definition of midpoint
$\angle ACM \cong \angle BCM$	Given
$CM = CM$	Reflexive property
$\triangle ACM \cong \triangle BCM$	**SAS**

Chapter 4 — Theorem Summary

#	Name	Key Idea
Thm 4-1	Angle Sum	$\angle A + \angle B + \angle C = 180^\circ$
Thm 4-2	Exterior Angle	Ext. $\angle$ = sum of remote interior $\angle$s
Cor 4-1	Third Angle	2 pairs congruent $\rightarrow$ 3rd pair congruent
Post 4-1	SSS	3 sides
Post 4-2	SAS	2 sides + included angle
Post 4-3	ASA	2 angles + included side
Thm 4-3	AAS	2 angles + non-included side
Thm 4-4	HL	Hypotenuse-leg (right triangles)
Thm 4-5	Isosceles	$\cong$ sides $\rightarrow$ $\cong$ base angles
Thm 4-6	Conv. Isosceles	$\cong$ angles $\rightarrow$ $\cong$ opposite sides
Cor 4-2	Equilateral	Equilateral $\iff$ equiangular (60° each)